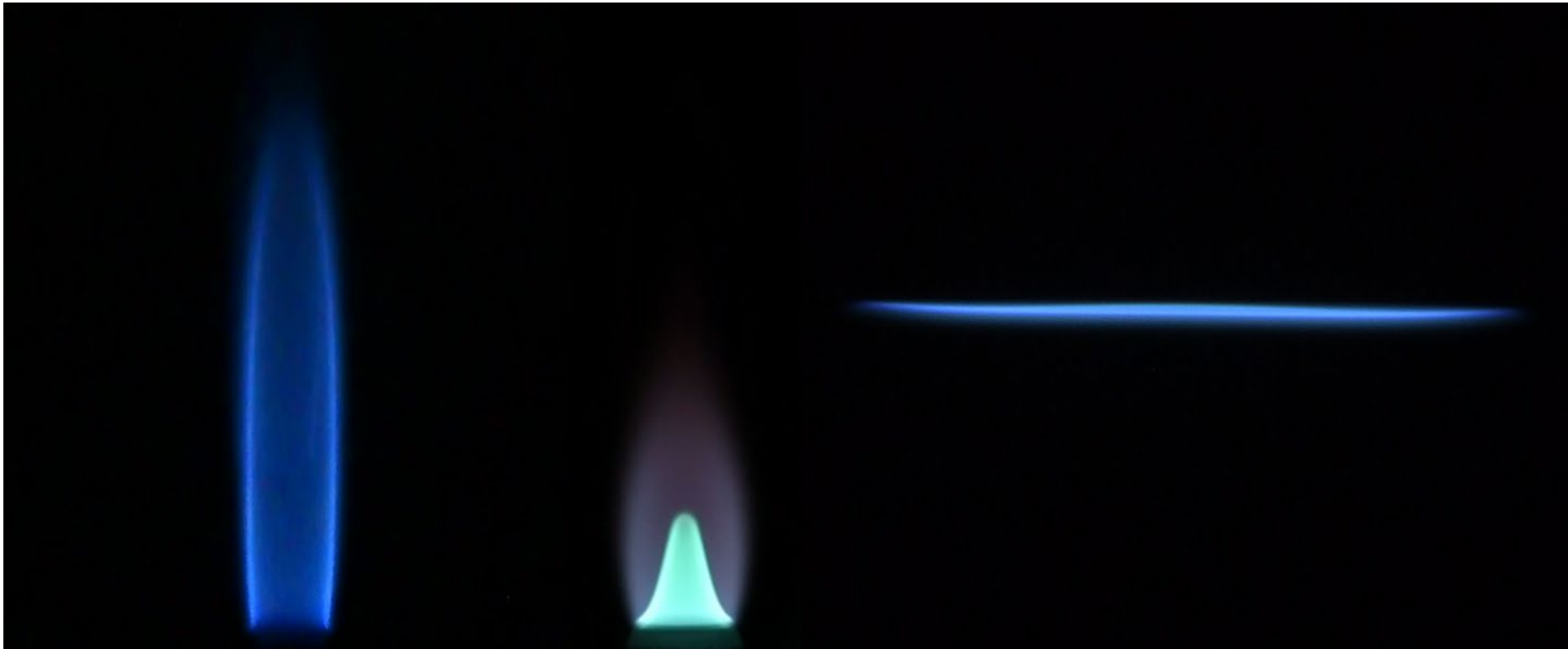


# 실습 가이드: OpenFOAM Laminar reacting flow

2026 한국연소학회 가을 Tutorial  
Next-Gen Combustion

# Laminar reacting flow

- Laminar reacting flow



# Turbulent reacting flow

## • Governing equations of multiComponentFluid

- A solver for transient, compressible, **turbulent** reacting flow included in the standard OpenFOAM framework.
- Here, the **unity Schmidt number** ( $Sc$ ) and **unity Lewis number** ( $Le$ ) assumptions are applied for simplification.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{v} Y_i) = \nabla \cdot (\mu_{\text{eff}} \nabla Y_i) + w_i$$

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) = \frac{\partial p}{\partial t} + \nabla \cdot (\alpha_{\text{eff}} \nabla h) + \sum h_i^0 w_i.$$

$$Sc = \frac{\nu}{D} = \frac{\text{kinematic viscosity}}{\text{mass diffusivity}}$$

$$Le = \frac{\alpha}{D} = \frac{\text{thermal diffusivity}}{\text{mass diffusivity}}$$

$$\mu_{\text{eff}} = \mu_l + \mu_t = \mu_l + \rho \nu_t$$

# Laminar reacting flow

## ● Governing equations of DTLreactingFoam solver [\*]

- A solver for transient, compressible, **laminar** reacting flow included in the DTLreactingFoam framework.
- Solves governing equations that account for species and energy transport due to **molecular diffusion**.
- Accurate evaluation of transport properties using a **detailed transport model (DTM)** is **essential**.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) + \nabla \cdot (\rho Y_i \mathbf{V}_i) = w_i$$

$$\begin{aligned} & \frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) - \frac{\partial p}{\partial t} \\ &= \nabla \cdot \left( \frac{\lambda}{c_p} \nabla h \right) - \nabla \cdot \left( \rho \sum_{i=1}^N h_i Y_i \mathbf{V}_i \right) + \sum h_i^0 w_i \end{aligned}$$

- Diffusion velocity

$$\mathbf{V}_i = -\frac{D_{i,\text{mix}}}{X_i} \nabla X_i$$

# Laminar reacting flow

- Detailed Transport Model (DTM) [\*]

- Transport properties such as viscosity  $\mu$ , thermal conductivity  $\lambda$ , and binary diffusivity  $D_{ji}$  are evaluated based on **standard kinetic theory** [\*]

- ❖ Viscosity

$$\mu_k = \frac{5}{16} \frac{\sqrt{\pi m_k k_B T}}{\pi \sigma_k^2 \Omega^{(2,2)*}}$$

- ❖ Thermal conductivity

$$\lambda_k = \frac{\mu_k}{m_k} (f_{\text{trans}} C_{v,\text{trans}} + f_{\text{rot}} C_{v,\text{rot}} + f_{\text{vib}} C_{v,\text{vib}})$$

- ❖ Binary diffusion coefficient

$$D_{ji} = \frac{3}{16} \frac{\left( \frac{2 \pi k_B^3 T^3}{m_{ji}} \right)^{1/2}}{p \pi \sigma_{ji}^2 \Omega^{(1,1)*}}$$

# Laminar reacting flow

- Detailed Transport Model (DTM) [\*]

- Mixture properties are obtained based on a **mixture-averaged model** [\*]

- ❖ Viscosity

$$\mu = \sum_{k=1}^K \frac{X_k \mu_k}{\sum_{j=1}^K X_j \Phi_{kj}} \quad \text{where} \quad \Phi_{kj} = \frac{1}{\sqrt{8}} \left( 1 + \frac{m_k}{m_j} \right)^{-\frac{1}{2}} \left( 1 + \left( \frac{\mu_k}{\mu_j} \right)^{\frac{1}{2}} \left( \frac{m_k}{m_j} \right)^{\frac{1}{4}} \right)^2$$

- ❖ Thermal conductivity

$$\lambda = \frac{1}{2} \left( \sum_{k=1}^K X_k \lambda_k + \frac{1}{\sum_{k=1}^K X_k / \lambda_k} \right)$$

- ❖ Mass diffusivity

$$D_{i,\text{mix}} = \frac{1 - Y_i}{\sum_{j \neq i}^K X_j / D_{ji}} \quad \text{where} \quad D_{ji} = \frac{3}{16} \frac{\left( \frac{2 \pi k_B^3 T^3}{m_{ji}} \right)^{1/2}}{p \pi \sigma_{ji}^2 \Omega^{(1,1)*}}$$

# Laminar reacting flow

## ● Existing framework for laminar reacting flow

- laminarSMOKE<sup>1</sup>
- OF-Cantera:
  - reactingDNS<sup>2</sup>,
  - CTreatingFoam<sup>3</sup>,
  - DeepFlame<sup>4</sup>
  - combustionFoam<sup>5</sup>
- reactingFoam-SCI<sup>6</sup>,
- laminarReactingFoam<sup>7</sup>
- standardReactingFoam<sup>8</sup>

- [1] A. Cuoci, A. Frassoldati, T. Faravelli, E. Ranzi, Combust. Flame 160 (2013) 870-886.
- [2] S. Zhong, F. Zhang, Z. Peng, F. Bai, Q. Du, Fuel 234 (2018) 1044-1054.
- [3] D. Zhou, H. Zhang, S. Yang, Aerospace 9 (2022).
- [4] R. Mao, M. Lin, Y. Zhang, T. Zhang, Z.-Q. J. Xu, Z. X. Chen, Comput. Phys. Commun. 291 (2023) 108842.
- [5] T. Li, J. Pan, F. Kong, B. Xu, X. Wang, Comput. Fluids 213 (2020) 104718.
- [6] Q. Yang, P. Zhao, H. Ge, Comput. Fluids 190 (2019) 114-127.
- [7] A. Dasgupta, E. Gonzalez-Juez, D. C. Haworth, Comput. Phys. Commun. 237 (2019) 219-229.
- [8] F. Wang, L. An, T. L. Chan, Comput. Phys. Commun. 309 (2025) 109472

# Laminar reacting flow

## ● Limitations of existing framework

### Advantage

- Provide a detailed transport model (DTM)
- Integrates with OF as a third-party package

### Disadvantage

- Does not fully **support all native** OF functionality
- The lack of a **user guide** and **C++ class diagrams** limits code extensibility and flexibility.

laminarSMOKE

### Advantage

- Provide detailed chemistry and transport through Cantera
- Integrates Cantera-based thermochemical models with OF

### Disadvantage

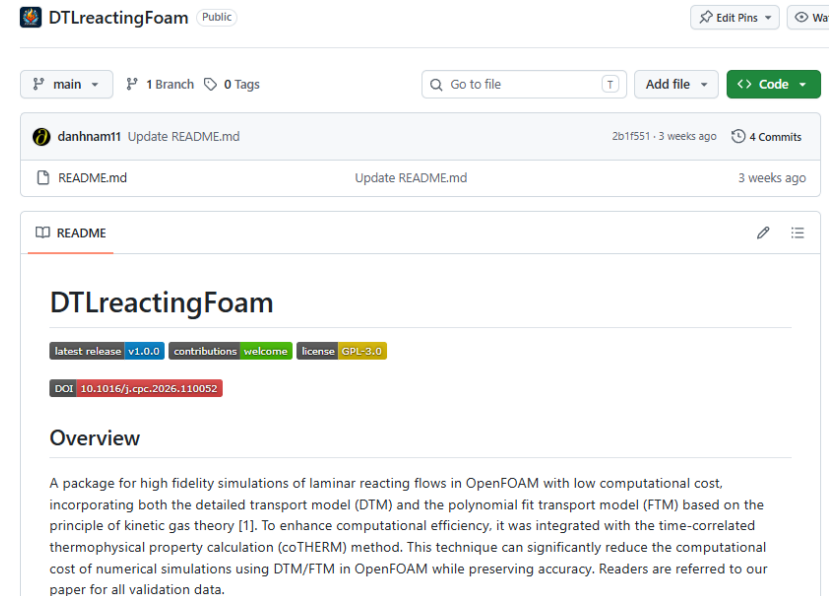
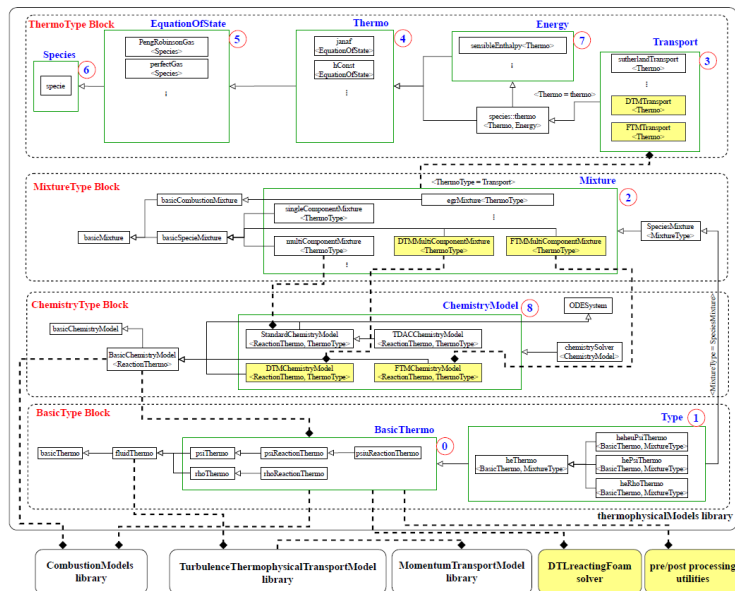
- **Dependency** on external library with **version compatibility** issues
- The **OF–Cantera interface** limits code extensibility and flexibility

Cantera coupled frameworks

# Laminar reacting flow

## ● DTLreactingFoam [\*]:

- Our group developed a **fully native** OF framework for DTM-based laminar reacting flow simulation while preserving **existing OF functionality**.
- Provides methods for reducing computational cost through **FTM** and **coTHERM** strategy.
- **Source code**, **User guide** and **tutorial cases** are available on GitHub repository [\*\*]



## ▪ Class diagram of DTLreactingFoam

## ▪ DTLreactingFoam GitHub repository [\*\*]

# Laminar reacting flow

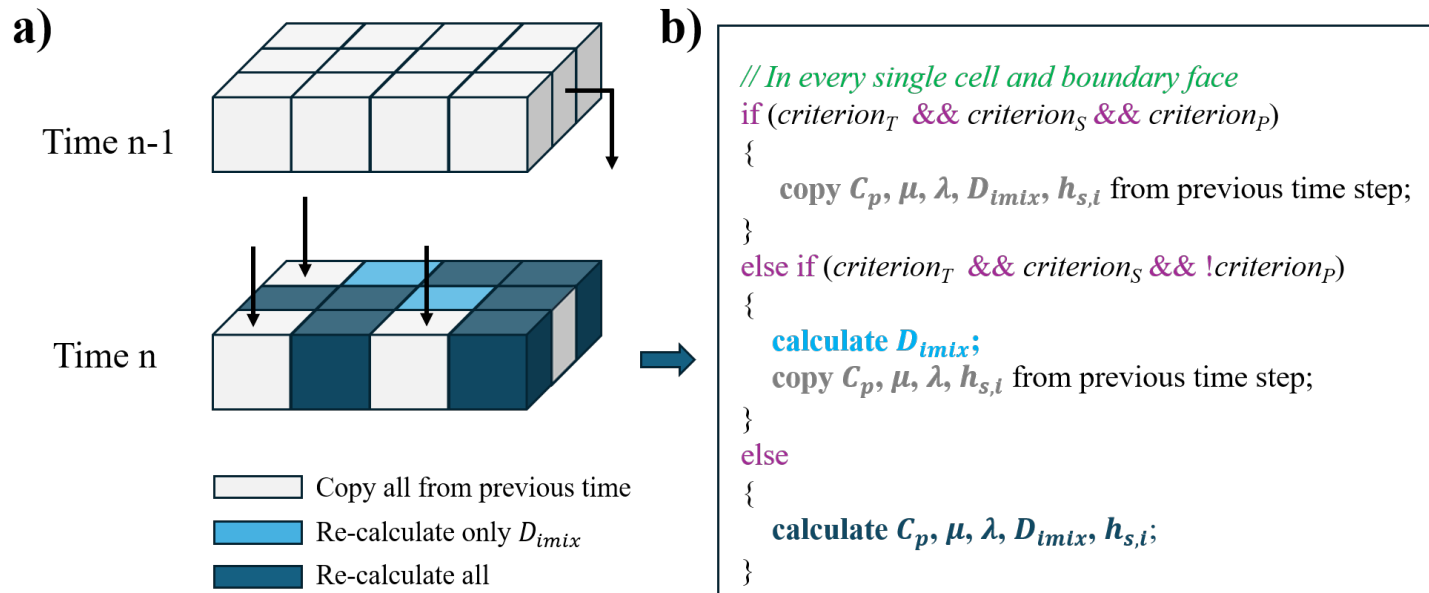
## ● Methods for reducing computational cost

### 1. Polynomial-fit transport models (FTM):

$$\ln \mu_k = \sum_{n=1}^N a_{n,k} (\ln T)^{n-1}; \quad \ln \lambda_k = \sum_{n=1}^N b_{n,k} (\ln T)^{n-1}; \quad \ln D_{jk} = \sum_{n=1}^N d_{n,jk} (\ln T)^{n-1}$$

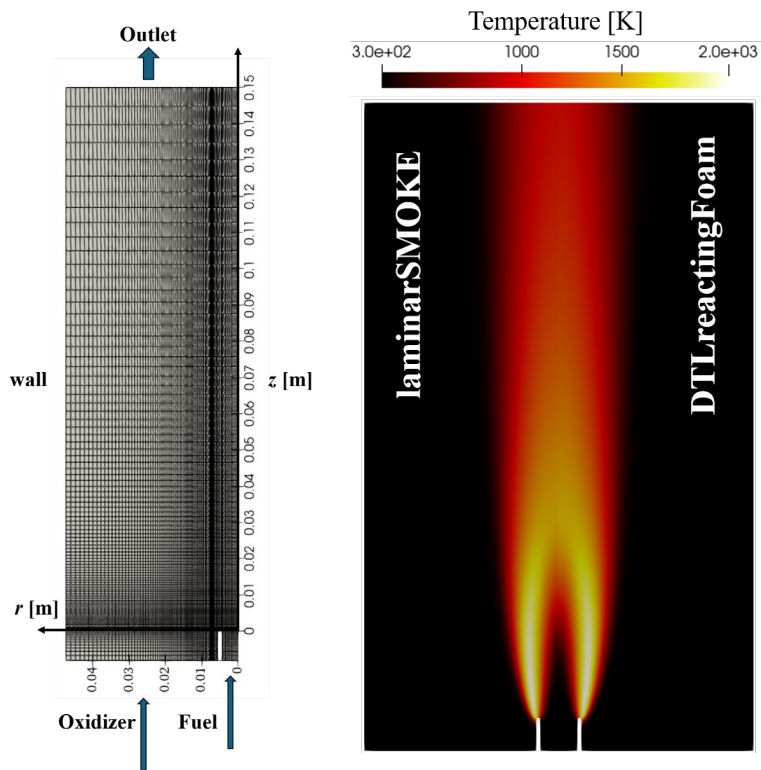
Using third-order polynomial fits:  $N = 4$

### 2. Time correlation for thermophysical properties (**coTHERM**): [\*]



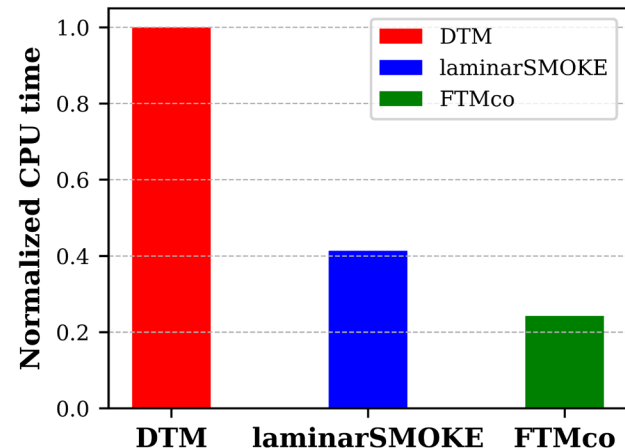
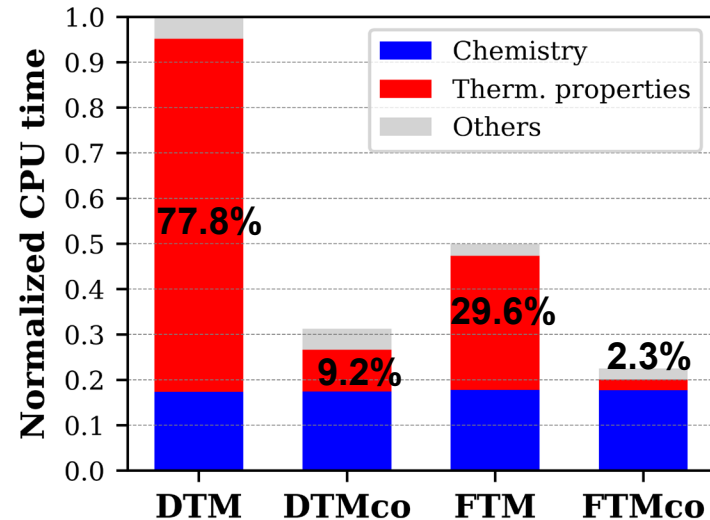
# Laminar reacting flow

## ● Computational efficiency of DTLreactingFoam



Threshold values

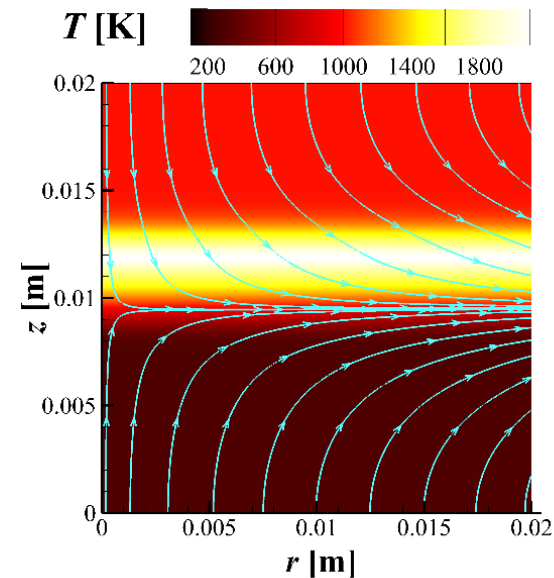
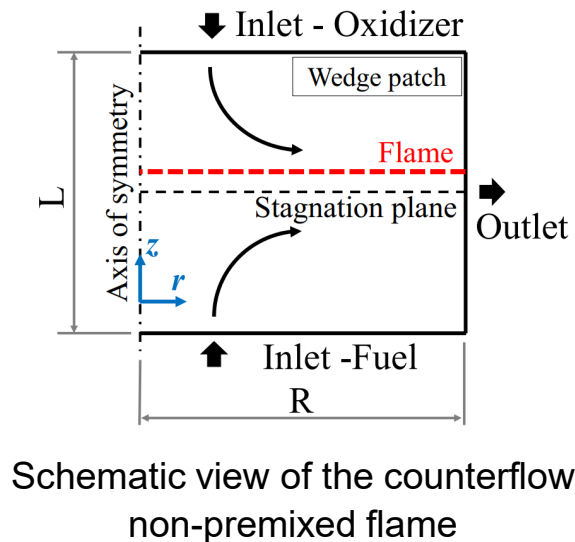
| $\varepsilon_T$ [K] | $\varepsilon_S$ [-] | $\varepsilon_p$ [Pa] |
|---------------------|---------------------|----------------------|
| 0.2                 | 1E-3                | 100                  |



# Laminar reacting flow

## ● Tutorial 3: Counter flow flame

- 2D **laminar** non-premixed counter flow flame
- Fuel: 50%  $\text{H}_2$  + 50%  $\text{N}_2$ , Oxidizer: 21%  $\text{O}_2$  + 79%  $\text{N}_2$
- Inlet velocity  $U_{\text{fuel}}$  : 0.5 m/s,  $U_{\text{oxi}}$  : 0.5 m/s, Pressure: 1 atm
- Perform simulations using **DTM** and **FTM + CoTHERM (FTMco)**.



The temperature isocontours of 2-D laminar counterflow non-premixed flame

---

# DTM model

# Laminar reacting flow

- Preprocessing

- Get the tutorial files

1

## Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
```

```
ghkrk@20225263-p01:~$ cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ ll
total 32
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x 1 ghkrk ghkrk  310 Oct  6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk  684 Oct  6 10:56 Allrun.DTM*
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 system/
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ |
```

# Laminar reacting flow

## ● Preprocessing: Kinetic mechanism

- Detailed hydrogen oxidation mechanism of Li et al. [\*].

**Table I** Detailed H<sub>2</sub>/O<sub>2</sub> Reaction Mechanism

|                                                                                                     | A                                | n     | E      | Reference |
|-----------------------------------------------------------------------------------------------------|----------------------------------|-------|--------|-----------|
| <b>H<sub>2</sub>/O<sub>2</sub> chain reactions</b>                                                  |                                  |       |        |           |
| 1. H + O <sub>2</sub> = O + OH                                                                      | $3.55 \times 10^{15}$            | -0.41 | 16.6   | [16]      |
| 2. O + H <sub>2</sub> = H + OH                                                                      | $5.08 \times 10^4$               | 2.67  | 6.29   | [44]      |
| 3. H <sub>2</sub> + OH = H <sub>2</sub> O + H                                                       | $2.16 \times 10^8$               | 1.51  | 3.43   | [45]      |
| 4. O + H <sub>2</sub> O = OH + OH                                                                   | $2.97 \times 10^6$               | 2.02  | 13.4   | [46]      |
| <b>H<sub>2</sub>/O<sub>2</sub> dissociation/recombination reactions</b>                             |                                  |       |        |           |
| 5. H <sub>2</sub> + M = H + H + M <sup>a</sup>                                                      | $4.58 \times 10^{19}$            | -1.40 | 104.38 | [47]      |
| H <sub>2</sub> + Ar = H + H + Ar                                                                    | $5.84 \times 10^{18}$            | -1.10 | 104.38 | [47]      |
| H <sub>2</sub> + He = H + H + He                                                                    | $5.84 \times 10^{18}$            | -1.10 | 104.38 | See text  |
| 6. O + O + M = O <sub>2</sub> + M <sup>a</sup>                                                      | $6.16 \times 10^{15}$            | -0.50 | 0.00   | [47]      |
| O + O + Ar = O <sub>2</sub> + Ar                                                                    | $1.89 \times 10^{13}$            | 0.00  | -1.79  | [47]      |
| O + O + He = O <sub>2</sub> + He                                                                    | $1.89 \times 10^{13}$            | 0.00  | -1.79  | See text  |
| 7. O + H + M = OH + M <sup>a</sup>                                                                  | $4.71 \times 10^{18}$            | -1.0  | 0.00   | [47]      |
| 8. H + OH + M = H <sub>2</sub> O + M <sup>b</sup>                                                   | $3.8 \times 10^{22}$             | -2.00 | 0.00   | See text  |
| <b>Formation and consumption of HO<sub>2</sub></b>                                                  |                                  |       |        |           |
| 9. H + O <sub>2</sub> + M = HO <sub>2</sub> + M <sup>c</sup>                                        | $k_0$ $6.37 \times 10^{20}$      | -1.72 | 0.52   | [19]      |
| H + O <sub>2</sub> + M = HO <sub>2</sub> + M <sup>d</sup>                                           | $k_0$ $9.04 \times 10^{19}$      | -1.50 | 0.49   | [19]      |
|                                                                                                     | $k_\infty$ $1.48 \times 10^{12}$ | 0.60  | 0.00   | [48]      |
| 10. HO <sub>2</sub> + H = H <sub>2</sub> + O <sub>2</sub>                                           | $1.66 \times 10^{13}$            | 0.00  | 0.82   | [1]       |
| 11. HO <sub>2</sub> + H = OH + OH                                                                   | $7.08 \times 10^{13}$            | 0.00  | 0.30   | [1]       |
| 12. HO <sub>2</sub> + O = OH + O <sub>2</sub>                                                       | $3.25 \times 10^{13}$            | 0.00  | 0.00   | [34]      |
| 13. HO <sub>2</sub> + OH = H <sub>2</sub> O + O <sub>2</sub>                                        | $2.89 \times 10^{13}$            | 0.00  | -0.50  | [34]      |
| <b>Formation and consumption of H<sub>2</sub>O<sub>2</sub></b>                                      |                                  |       |        |           |
| 14. HO <sub>2</sub> + HO <sub>2</sub> = H <sub>2</sub> O <sub>2</sub> + O <sub>2</sub> <sup>e</sup> | $4.20 \times 10^{14}$            | 0.00  | 11.98  | [49]      |
| HO <sub>2</sub> + HO <sub>2</sub> = H <sub>2</sub> O <sub>2</sub> + O <sub>2</sub>                  | $1.30 \times 10^{11}$            | 0.00  | -1.63  |           |
| 15. H <sub>2</sub> O <sub>2</sub> + M = OH + OH + M <sup>f</sup>                                    | $k_0$ $1.20 \times 10^{17}$      | 0.00  | 45.5   | [50]      |
|                                                                                                     | $k_\infty$ $2.95 \times 10^{14}$ | 0.00  | 48.4   | [51]      |
| 16. H <sub>2</sub> O <sub>2</sub> + H = H <sub>2</sub> O + OH                                       | $2.41 \times 10^{13}$            | 0.00  | 3.97   | [47]      |
| 17. H <sub>2</sub> O <sub>2</sub> + H = H <sub>2</sub> + HO <sub>2</sub>                            | $4.82 \times 10^{13}$            | 0.00  | 7.95   | [47]      |
| 18. H <sub>2</sub> O <sub>2</sub> + O = OH + HO <sub>2</sub>                                        | $9.55 \times 10^6$               | 2.00  | 3.97   | [47]      |
| 19. H <sub>2</sub> O <sub>2</sub> + OH = H <sub>2</sub> O + HO <sub>2</sub> <sup>e</sup>            | $1.00 \times 10^{12}$            | 0.00  | 0.00   | [52]      |
| H <sub>2</sub> O <sub>2</sub> + OH = H <sub>2</sub> O + HO <sub>2</sub>                             | $5.8 \times 10^{14}$             | 0.00  | 9.56   |           |

In tran.dat:  
Transport parameters for DTM

| ! Species                                                | ! Geometry | ! epsilon/kB | ! sigma | ! Dipole | ! Polarizability | ! Zrot      |
|----------------------------------------------------------|------------|--------------|---------|----------|------------------|-------------|
| ! Geometry : 0 = monatomic, 1 = linear, 2 = nonlinear    |            |              |         |          |                  |             |
| ! epsilon/kB : Lennard-Jones well depth [K]              |            |              |         |          |                  |             |
| ! sigma : Lennard-Jones diameter [Angstrom]              |            |              |         |          |                  |             |
| ! Dipole : Dipole moment [Debye]                         |            |              |         |          |                  |             |
| ! Polarizability : [Angstrom^3]                          |            |              |         |          |                  |             |
| ! Zrot : Rotational relaxation collision number at 298 K |            |              |         |          |                  |             |
| AR                                                       | 0          | 136.500      | 3.330   | 0.000    | 0.000            | 0.000       |
| AS                                                       | 0          | 1045.500     | 4.580   | 0.000    | 0.000            | 0.000 ! MEC |
| ASH                                                      | 1          | 199.300      | 4.215   | 0.000    | 0.000            | 1.000 ! MEC |
| ASH2                                                     | 2          | 229.600      | 4.180   | 0.000    | 0.000            | 1.000 ! MEC |
| C                                                        | 0          | 71.400       | 3.298   | 0.000    | 0.000            | 0.000 ! *   |
| C2                                                       | 1          | 97.530       | 3.621   | 0.000    | 1.760            | 4.000       |
| C2O                                                      | 1          | 232.400      | 3.828   | 0.000    | 0.000            | 1.000 ! *   |
| CN2                                                      | 1          | 232.400      | 3.828   | 0.000    | 0.000            | 1.000 ! OIS |
| C2H                                                      | 1          | 265.300      | 3.721   | 0.000    | 0.000            | 2.500 ! NMM |
| C2H2                                                     | 1          | 265.300      | 3.721   | 0.000    | 0.000            | 2.500 ! NMM |

# Laminar reacting flow

## ● Preprocessing: Kinetic mechanism

- Convert CHEMKIN-format chemistry, thermodynamic, and transport data into OpenFOAM-compatible DTM thermodynamic and reaction file

②

### Command Line

```
DTMchemkinToFoam \  
  chemkin/chem.inp \  
  chemkin/therm.dat \  
  chemkin/tran.dat \  
  chemkin/transportSutherland \  
  constant/reactions \  
  constant/thermo.DTM
```

Input

Output

# Laminar reacting flow

## ● Preprocessing: Kinetic mechanism

- Convert Chemkin-format files into OF-format reaction and and thermodynamic file

3

### Command Line

```
vi constant/thermo.DTM
```

To exit vi editor, type **:q**

|            |                                   |
|------------|-----------------------------------|
| <b>:wq</b> | write and quit the file           |
| <b>:w</b>  | write                             |
| <b>:q</b>  | quit the file                     |
| <b>:q!</b> | quit the file (ignore the change) |

```
species          9 ( H2 O2 O OH H2O H HO2 H2O2 N2 );

OH
{
  specie
  {
    molWeight      17.00737;
  }
  thermodynamics
  {
    Tlow           200;
    Thigh          6000;
    Tcommon        1000;
    highCpCoeffs   ( 2.86472886 0.00105650448 -2.59082758e-07 3.05218674e-11 -1.33195876e-15 3683.62875 5.70164073 );
    lowCpCoeffs    ( 4.12530561 -0.00322544939 6.52764691e-06 -5.79853643e-09 2.06237379e-12 3346.30913 -0.69043296 );
  }
}

transport
{
  As              1.512e-06;
  Ts              120;
  linearity       1;
  epsilonOverKb   80;
  sigma           2.75;
  dipoleMoment    0;
  alpha           0;
  Zrot            0;
}

elements
{
  O              1;
  H              1;
}
}
```

Transport parameters for DTM

# Laminar reacting flow

## ● Preprocessing: Physical models

1. Enable detailed transport and configure CoTHERM algorithm.
2. Set the CoTHERM threshold for  $T$ ,  $p$ , and  $Y_i$ .
3. Select the multicomponent mixture and detailed transport models.
4. Include thermo.DTM.

4

### Command Line

```
vi constant/physicalProperties
```

```
/*----- C++ -----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location     "constant";
    object       physicalProperties;
}
// *****

usingDetailedTransportModel true;
usingCoTHERM false;
usingPreProcessingFTM false;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT 0.2; // [K]
epsilonP 100; // [Pa]
epsilonS 0.001; // [-], in mass fraction

thermoType
{
    type hePsiThermo;
    mixture DTMMulticomponentMixture;
    transport DTMTransport;
    thermo janaf;
    energy sensibleEnthalpy;
    equationOfState perfectGas;
    specie specie;
}

defaultSpecie N2;

#include "thermo.DTM"

// *****
```

1

2

3

4

# Laminar reacting flow

- Preprocessing: Mesh Generation

5

## Command Line

blockMesh

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ blockMesh
/*-----*/
=====
\\  F ield      | OpenFOAM: The Open Source CFD Toolbox
\\  O peration  | Website:  https://openfoam.org
\\  A nd        | Version:  12
\\  M anipulation|
/*-----*/

Build   : 12-86e126a7bc4d
Exec    : blockMesh
Date    : Oct 06 2026
Time    : 11:01:29
Host    : "20225263-p01"
PID     : 73088
I/O     : uncollated
Case    : /home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
nProcs  : 1
sigFpe  : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// *****
Create time
```

# Laminar reacting flow

- Preprocessing: Domain decomposition
  - Parallel computing

In system/decomposeParDict file

```
/*-----* C++ *-----*/
//
// \ \ \ \ \
//  F ield      | OpenFOAM: The Open Source CFD Toolbox
//  O peration  | Website: https://openfoam.org
//  A nd        | Version: 12
//  M anipulation|
//
/*-----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    note        "mesh decomposition control dictionary";
    object      decomposeParDict;
}
// ***** //

numberOfSubdomains 4;

method          simple;

simpleCoeffs
{
    n             (1 4 1);
    delta         0.001;
}
// ***** //
```

6

## Command Line

```
vi system/decomposeParDict
```

# Laminar reacting flow

- Preprocessing: Domain decomposition
  - Parallel computing

7

## Command Line

```
decomposePar
```

```
11
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ 11
total 52
drwxr-xr-x 10 ghkrk ghkrk 4096 Oct  7 16:56 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  6 14:18 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x  1 ghkrk ghkrk  310 Oct  6 10:56 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  684 Oct  6 10:56 Allrun*
-rw-r--r--  1 ghkrk ghkrk 2156 Oct  6 14:54 README.md
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  7 16:56 constant/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor0/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor1/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor2/
drwxr-xr-x  4 ghkrk ghkrk 4096 Oct  7 16:56 processor3/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  6 11:03 system/
```

Domain is decomposed

# Laminar reacting flow

- Preprocessing: Simulation control

8

## Command Line

```
vi system/controlDict
```

```
/*----- C++ -----*/
//
//   F i e l d
//   O p e r a t i o n
//   A n d
//   M a n i p u l a t i o n
//
//   OpenFOAM: The Open Source CFD Toolbox
//   Website:  https://openfoam.org
//   Version:  12
//
/*-----*/

FoamFile
{
    format      ascii;
    class       dictionary;
    location    "system";
    object      controlDict;
}
// ***** //

application    foamRun;
solver          DTLreactingFoam;
startFrom      startTime;
startTime      0;
stopAt         endTime;
endTime        0.1;
deltaT         1e-6;
writeControl    adjustableRunTime;
writeInterval   0.005;
purgeWrite     0;
writeFormat     ascii;
writePrecision  6;
writeCompression off;
timeFormat      general;
timePrecision   6;
runtimeModifiable true;
adjustTimeStep  yes;
```

# Laminar reacting flow

- Solving: Run the simulation
  - Use DTLreactingFoam solver

⑨

## Command Line

```
mpirun -np 4 foamRun -parallel
```

# Laminar reacting flow

## ● Post-Processing

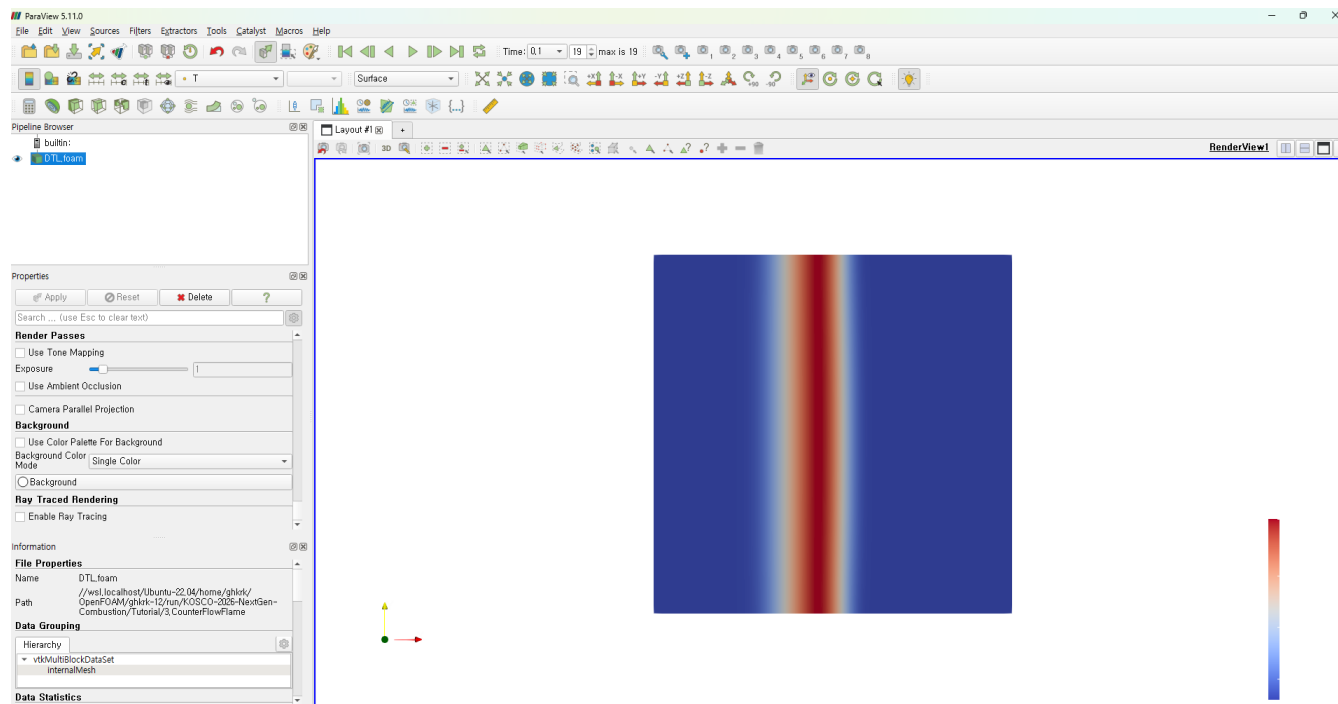
### ⑩ Command Line

```
reconstructPar
```

```
touch DTM.foam
```

→ Copy & paste files  
to C: or D: drive

→ Open foam file  
with paraview



---

# **FTM model with coTHERM (FTMco)**

# Laminar reacting flow

- Preprocessing for FTM

- Get the tutorial files

1

## Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTMC0
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTMC0$ ll
total 36
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 14:55 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 14:18 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x 1 ghkrk ghkrk  328 Oct  6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk 2451 Oct  6 14:53 Allrun*
-rw-r--r-- 1 ghkrk ghkrk 1146 Oct  6 14:55 README.md
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 14:54 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 system/
```

# Laminar reacting flow

- **Preprocessing for FTM: Polynomial coefficient generation**

- Enable FTM preprocessing.
- Set the transport model to `preprocessingFTMTransport`.

```

/*-----*- C++ -*/
=====
\\      F ield          OpenFOAM: The Open Source CFD Toolbox
\\      O peration       Website: https://openfoam.org
\\      A nd              Version: 12
\\      M anipulation

/*-----*/

FoamFile
{
    format        ascii;
    class         dictionary;
    location      "constant";
    object        physicalProperties;
}

// ***** //

usingDetailedTransportModel true;
usingCoTHERM false;
usingPreProcessingFTM true;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT           0.2; // [K]
epsilonP           100; // [Pa]
epsilonS           0.001; // [-], in mass fraction

thermoType
{
    type            hePsiThermo;
    mixture          DTMMulticomponentMixture;
    transport        preprocessingFTMTransport
    thermo           janaf;
    energy           sensibleEnthalpy;
    equationOfState perfectGas;
    specie           specie;
}

defaultSpecie N2;

#include "thermo.DTM"

// ***** //

```

## 2 Command Line

```
vi constant/physicalProperties
```

# Laminar reacting flow

- Preprocessing for FTM: Polynomial coefficient generation
  - Generate thermo.FTM containing the FTM transport coefficients.

3

## Command Line

```
blockMesh
```

```
FTMchemkinToFoam
```

```
vi constant/thermo.FTM
```

```
OH
{
  transport
  {
    muCoeffs      (-17.3099 1.90275 -0.170195 0.00767127);
    kappaCoeffs   (4.12383 -3.89714 0.629982 -0.0280239);
    DijCoeffs     (
      (-20.112 2.28781 -0.0847962 0.00377877)
      (-23.4526 3.14265 -0.197273 0.00872065)
      (-22.376 2.88806 -0.164634 0.00732455)
      (-9.21034 -6.44736e-26 9.25847e-27 -4.40439e-28)
      (-28.6142 5.15624 -0.446304 0.0190235)
      (-22.5274 3.44613 -0.235979 0.010369)
      (-23.4579 3.14265 -0.197273 0.00872065)
      (-23.463 3.14265 -0.197273 0.00872065)
      (-23.2325 3.04984 -0.185302 0.00820502)
    );
  }
}
```

# Laminar reacting flow

## ● Preprocessing for FTM

- Disable FTM preprocessing and enable CoTHERM.
- Select FTMMulticomponentMixture and FTMTransport as the mixture and transport models.
- Include thermo.FTM containing the generated transport coefficients.

4

### Command Line

```
vi constant/physicalProperties
```

```
/*----- C++ -----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location     "constant";
    object       physicalProperties;
}
// *****

usingDetailedTransportModel true;
usingCoTHERM true;
usingPreProcessingFTM false;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT      0.2; // [K]
epsilonP      100; // [Pa]
epsilonS      0.001; // [-], in mass fraction

thermoType
{
    type        hePsiThermo;
    mixture      FTMMulticomponentMixture;
    transport     FTMTransport;
    thermo       janaf;
    energy        sensibleEnthalpy;
    equationOfState perfectGas;
    specie        specie;
}

defaultSpecie N2;

#include "thermo.FTM"

// *****
```

# Laminar reacting flow

- **Solving: Run the simulation with FTM**
  - Use DTLreactingFoam solver

5

## Command Line

```
decomposePar
```

```
mpirun -np 4 foamRun -parallel
```

# Laminar reacting flow

## ● Post-Processing

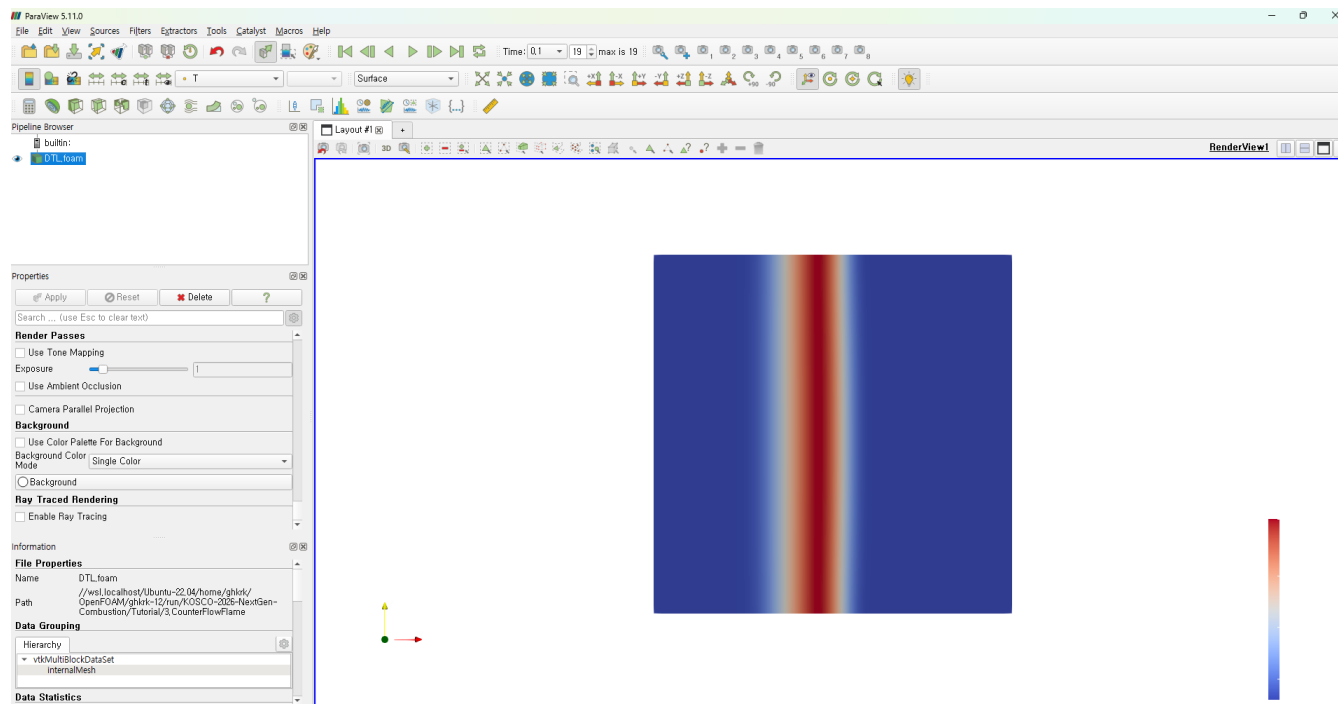
### ⑩ Command Line

```
reconstructPar
```

```
touch FTMC0.foam
```

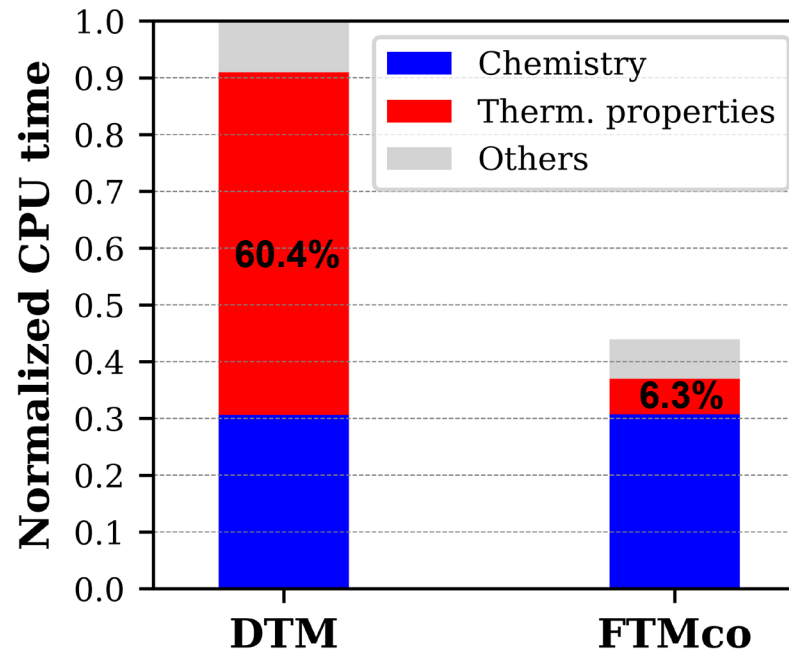
→ Copy & paste files  
to C: or D: drive

→ Open foam file  
with paraview



# Laminar reacting flow

- Computational efficiency for CounterFlowFlame Case



CPU time: 56%  
reduction